

Athirath Bommerla

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PROFESSIONAL SUMMARY

Data Engineer combining data engineering and analytics to turn complex operational/field data into reliability insights. Builds scalable ETL pipelines, curated datasets, and dashboards using Python, SQL, Airflow, Spark, and AWS to surface failure trends and support prioritization and root-cause analysis.

TECHNICAL SKILLS

Programming & Analysis: Python, SQL, dbt, Pandas, NumPy, DuckDB, Statistical Analysis

Data Engineering: Airflow, ETL/ELT Pipelines, Incremental Loads, Schema Validation, API Integrations

Big Data & Compute: Spark; Hadoop & HDFS (concepts + hands-on file operations), Hive basics

Dashboards & Visualization: Tableau, Power BI, Matplotlib, React.js, Plotly

Backend Development: .NET, FastAPI, C#, REST APIs

DevOps & Platforms: Git, AWS (S3, Lambda), Docker, Kubernetes fundamentals

Observability: Splunk-style log analysis, metrics-driven monitoring, alerting workflows

Domain Knowledge: Reliability Analytics, Fleet/Field Trend Analysis, Manufacturing & Facilities Operations, SCADA (Ignition)

PROFESSIONAL EXPERIENCE

Toyota Motors North America

Plano, TX

Data Engineer

Feb 2026 – Jun 2026

- Diagnosed TRI (Total Recordable Incident) reporting anomalies across ingestion pipelines where safety data lacked traceability, restoring accurate compliance metrics and eliminating manual reconciliation for safety teams.
- Rebuilt broken VEADS (Vehicle Engineering Analytical Data System) pipelines producing inconsistent R&D datasets, unifying cross-functional engineering data and making it reliable for product development analytics.
- Identified data reliability gaps inside Enclave, Toyota's internal secure deployment platform, and engineered validation and monitoring layers so customers could interact with enterprise tools on governed, trustworthy data.
- Streamlined redundant pipeline architecture and automated validation checks that were inflating operational costs, delivering approximately 20–25% cost reduction across the data engineering function.

University of Oklahoma Health Sciences Center / DISC

Norman, OK

Graduate Research & Data Engineer

Aug 2024 – Jun 2025

- Collected and translated complex stakeholder logic into maintainable implementations, building Python + SQL data transformations that enabled consistent decision-making under changing operational requirements.
- Designed and maintained end-to-end ETL pipelines using Apache Airflow to ingest multi-source telemetry and operational signals, producing analytics-ready datasets with robust retry/backfill patterns.
- Performed reliability-style analyses by detecting failure patterns, anomaly trends, and recurring incident signatures across equipment and sensor streams, helping prioritize corrective actions for operators and engineers.
- Automated incident triage workflows using SQL-driven rules + Python validators, improving auditability, reducing manual analysis, and enabling repeatable reprocessing during investigations.

Indeyes Technologies

Hyderabad, India

Data Engineer Intern

May 2022 – Dec 2022

- Built and optimized ETL/ELT pipelines to consolidate legacy operational datasets into governed SQL data marts, applying incremental refresh patterns and standardized transformation logic.
- Developed Power BI/Tableau-style KPI reporting backed by optimized SQL models, improving trend visibility and reducing ad-hoc analysis time for operations stakeholders.
- Implemented cross-system API integrations with validation and error-handling to support near real-time updates and reliable downstream consumption.
- Partnered with engineering stakeholders to translate ambiguous requirements into simple deliverables, productionizing analytics workflows with modular code, documentation, and version-controlled releases.

SELECTED PROJECTS

- Facilities/Fleet Reliability Data Pipeline** | *Python, Airflow, AWS, dbt, SQL, Tableau/Power BI* 2025
- Designed an end-to-end pipeline with Airflow to ingest telemetry-style datasets and publish curated tables for trend analysis, reliability KPIs, and prioritization.
 - Implemented dbt models to standardize schemas and compute metrics such as uptime, incident rate, and energy efficiency with reproducible transformations.
 - Added data quality checks (constraints, drift detection, reconciliation) and fault-tolerant retries/backfills to improve robustness under evolving source formats.
 - Built dashboards in Power BI/Tableau-style layers to visualize failure trends and system health for rapid investigation and reporting.
- VADE — Voice-Activated Data Engineer** | *Python, Dagster, Pandera, DuckDB, SpeechRecognition* 2025
- Developed a voice-controlled orchestration system using Dagster to trigger ETL jobs and monitor pipeline execution states.
 - Built schema validation layers in Pandera to automatically detect, log, and surface anomalies that impact downstream analytics and reporting.
 - Stored structured logs and workflow states in DuckDB to enable sub-second querying for performance and reliability diagnostics.
 - Enabled voice-based execution of validation, load, and monitoring commands to emulate operator-style workflows and rapid triage.
- NeuroSync — AI Productivity Assistant** | *Next.js, FastAPI, Supabase, Qdrant, LangChain* 2025
- Created a semantic search platform for operational and maintenance logs using Qdrant vector embeddings to accelerate investigation workflows.
 - Implemented REST APIs in FastAPI for ingestion, summarization, and retrieval to support rapid root-cause exploration and reporting.
 - Integrated Supabase authentication and RLS policies for secure role-based access to internal datasets and dashboards.
 - Built UI workflows for trend detection, incident summarization, and knowledge retrieval to reduce time-to-insight for engineers.
- DashClone — Operational Analytics Dashboard** | *React.js, Supabase, Tailwind, shadcn/ui, Node.js* 2025
- Built a real-time operational dashboard to visualize asset usage, event-driven KPIs, and system health signals for monitoring and decision support.
 - Configured Supabase backend with role-based access control and real-time listeners for near real-time updates across datasets.
 - Developed APIs in Node.js for aggregation and anomaly flagging, supporting reliability-style trend identification.
 - Optimized rendering and query execution to achieve 40% faster dashboard load times and improved reporting responsiveness.

EDUCATION

- University of Oklahoma** Norman, OK
M.S. Data Science and Analytics — GPA 3.55/4.0 2023 – 2025
- Sreenidhi Institute of Science and Technology** Hyderabad, India
B.Tech Computer Science — CGPA 3.85/4.0 2019 – 2023